



We're Hungry

More than 25% of all ice-free land on this Earth is taken up by livestock. Those are some really hungry creatures indeed.



It's really windy today

Did you know that winds on Neptune can touch 2,200km/hr? Imagine wind surfing in that lot!

PHOTO BY: SPL/ISTOCK



A tardigrade specimen, commonly known as a water bear

survival reality TV shows seem dull in comparison. Let's start with a fun fact – tardigrades are the first animals that could survive the dangerous vacuum of space. A bunch of these creatures orbited in outer space on the outside of a FOTON-M3 rocket. They were bombarded with all

DID YOU KNOW?

The superpowers of the tardigrade family were discovered way back in 1770's. In 1995, dried tardigrades were brought back to life, after eight years of water deprivation!

sorts of cosmic radiations and plenty of them made the trip back home to tell us their story!

Not only are these organisms surprisingly suited for life in space, they can also withstand temperatures ranging from just above absolute zero to way above the boiling point of water. Also, sample this: they can survive vacuum as well as the pressure at Mariana Trench – 11 kilometres deep in the Pacific Ocean.

Studies have attributed some of these amazing powers to the tardigrades' ability to practice forms of cryptobiosis – a state in which metabolic activity is severely slowed down. This state, called desiccation, allows the creature to lose water and almost arrest all metabolism. When blessed with water, the tardigrade simply regains its rest state and moves on in life as if nothing happened. Sure, this ability can help it survive deserts and droughts, but how does it help this water bear survive in space or extreme temperatures?

In its desiccated form, the tardigrade performs some life-saving functions. A

sugar molecule prohibits cellular expansion and anti-oxidants produced neutralise the threat posed by oxygen-reacting molecules present in outer space radiation. The anti-oxidants can help repair damaged DNA and this ability could explain the tardigrades' knack for surviving extreme pressures. While all these functions present clues to the secret of the tardigrade's superpowers, we understand very little about how the tardigrade functions on a molecular level. There's little evidence about its evolutionary history as well. Does its flair for outer space survival point to an extraterrestrial origin?

Research about tardigrades can have exciting implications. If cryonics became feasible, think of the applications! Medicines and tablets could be stored at room temperature and perhaps, we could design a super-suit designed for extreme exploration. Astrobiologists will be able to stretch their parameters to search for life outside Earth. If a microorganism on Earth could test nature's limit to such an extent, maybe there are such organisms on places like Jupiter's moons, lying in an unactivated state, just waiting to be discovered.

Dark energy and Dark matter

The study of matter on Earth is an all-encompassing field. It's a pity that all matter known to us makes up just about 5% of the known universe! The rest of the universe is "dark" and its major constituents are aptly named 'dark matter' (~27%) and 'dark energy' (~68%).

Any list of unsolved problems in science is incomplete without the mention of the mystery of dark matter and dark energy. Dark energy is the proposed reason for the acceleration of the universe. In 1998, when two different research teams confirmed that the expansion of the universe was accelerating, it debunked the then popular notion of gravity slowing

down the universe's expansion. Theorists had, and still have, a headache explaining this, and in grand fashion named the possible explanation as dark energy. So far, no real contender has emerged for a correct explanation of the form dark energy takes. Current solutions state that dark energy could be a property of space and lend empty space energy, or it could be a sort of space-filling fluid that somehow leads to the acceleration of the universe, in the way that 'normal' energy cannot.

Dark matter is weird. It interacts with almost nothing, not even light, making its detection very difficult. Dark matter was discovered when researchers found oddities in the dynamics of certain galaxies. The known mass of the galaxy couldn't explain the deviations and they concluded that some form of invisible matter with gravitational pull existed throughout the universe. Dark matter has never been directly detected, but scientists have however observed the effects of dark matter via gravitational lensing (bending of light due to gravitational interaction with invisible matter). Thank god for dark matter's gravitational pull!

The constituent of dark matter is one of the biggest problems in particle physics and cosmology today. Scientists are of the opinion that dark matter is made up of certain exotic particles called 'WIMP' (or 'Weakly Interacting Massive Particles') that owe their existence to a theory called 'Supersymmetry'. Scientists also postulate that dark matter could be made up of baryons. Baryonic dark matter could be 'MACHOS' (or 'Massive Astrophysical Compact Halo Objects') like small black holes.

While both theories – dark matter and dark energy – arise from our inability to explain some observed features of the universe, they're in essence competing forces of the cosmos and both attract massive funding for large scale experiments. Dark energy repels and dark matter attracts. The dominance of either force will decide if the universe will expand forever or be drawn towards itself and ultimately implode in a rather interesting, Big Bang-type event, christened the "Big Crunch". For now, however, we're truly in the dark about both theories. 

DID YOU KNOW?

A hypothetical form of dark energy called 'phantom dark energy' exists, which can cause such rapid expansion of the universe that everything will disintegrate into its elementary constituents and radiation. The event has been aptly named "Big Rip".